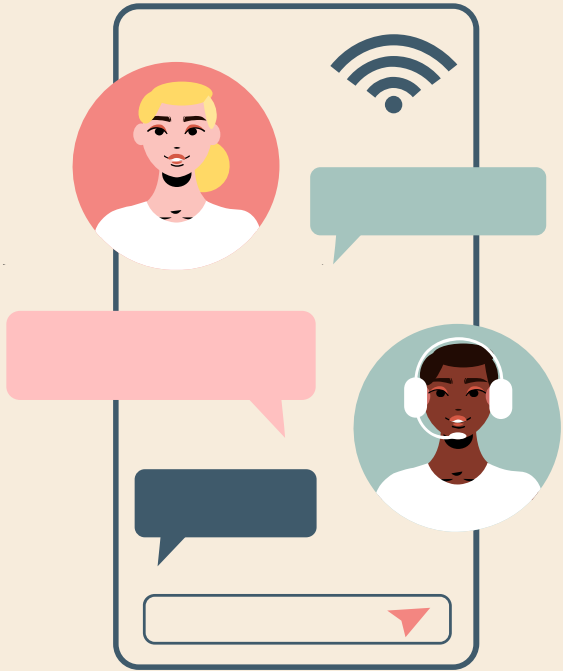


Data Analysis – Use Case II: Customer Support Data

Identifying Trends and Driving Improvements

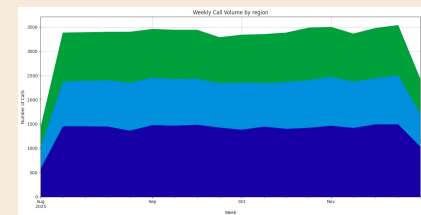
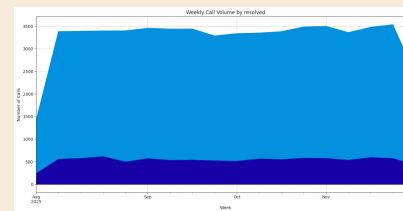
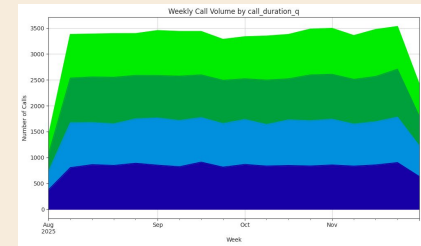
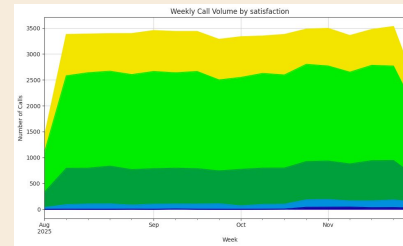
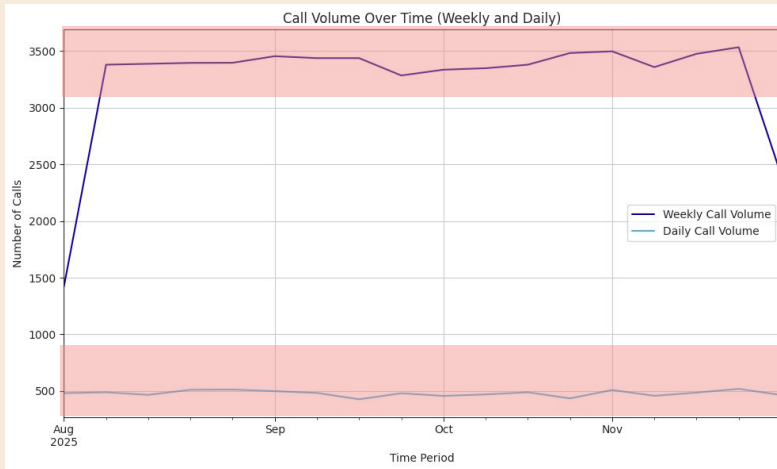
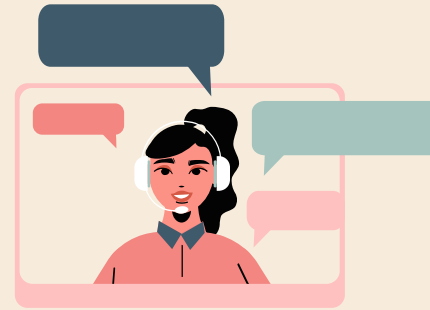


Mati Borlak, Brenna Hymowitz, Aviv Goldshide



Dataset Overview

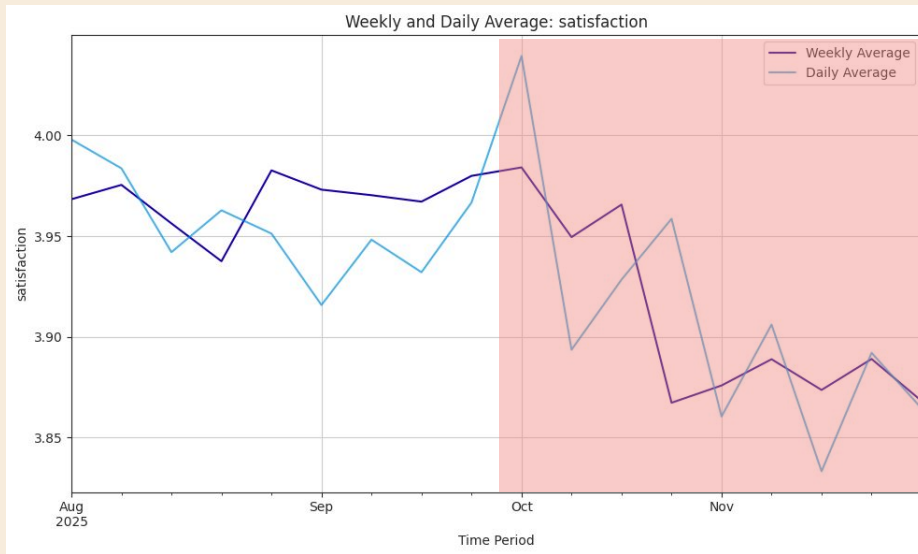
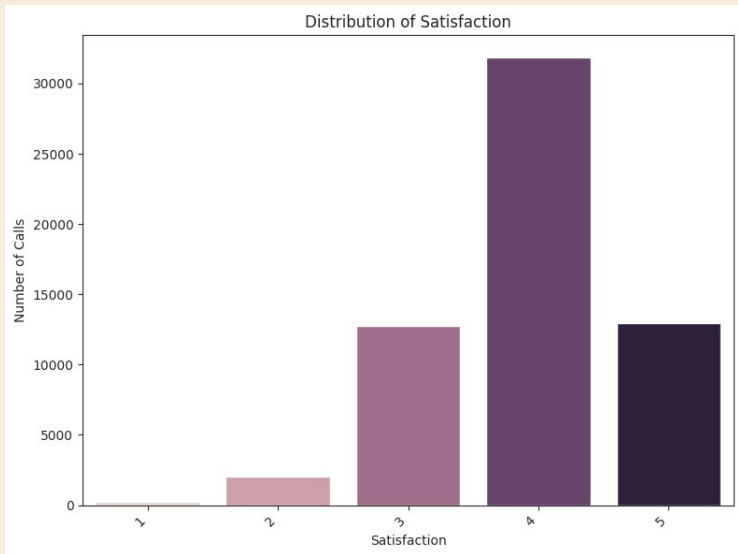
- 60,000 customer support calls from August to November 2025
- Call and customer demographics
- Steady (flat distribution) call volume



Customer Satisfaction



- Customer **satisfaction** is crucial, reflecting the effectiveness of our service and impacting reputation and loyalty (churn)
- Average customer **satisfaction** drops from October to November

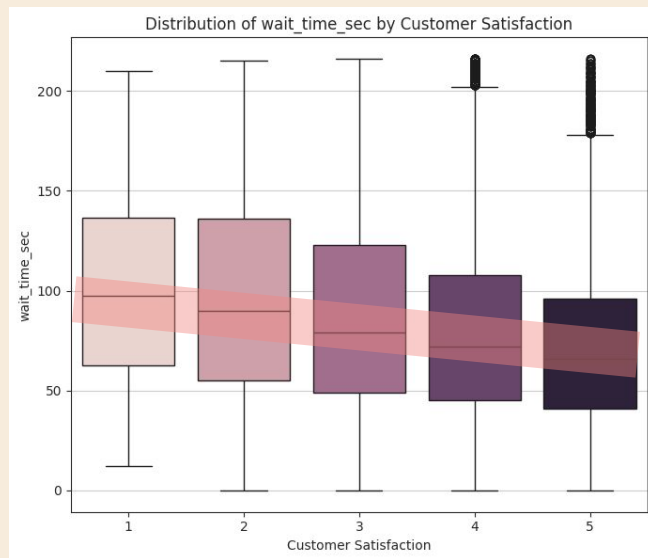
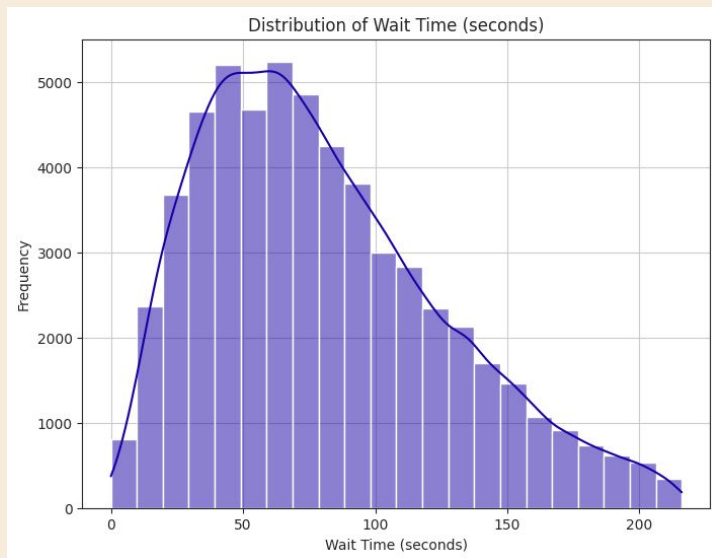


Missing Data

- Missing **wait time** values were primarily from calls with the lowest customer **satisfaction**
- Longer **wait times** lead to lower **satisfaction**

wait_time_sec	2517
call_duration_sec	0
issue_type	192

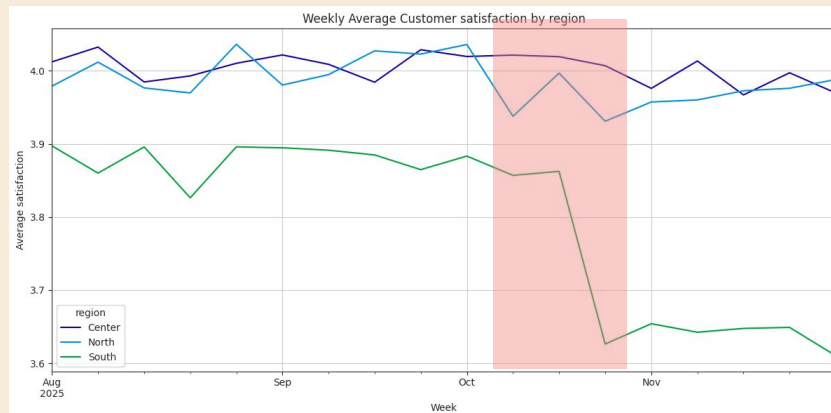
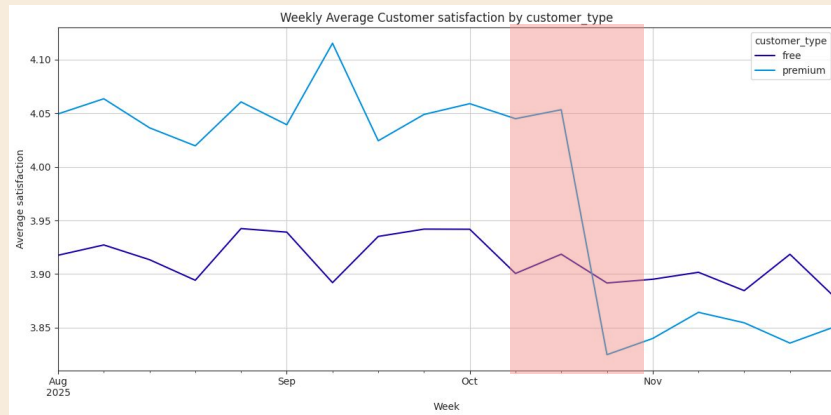
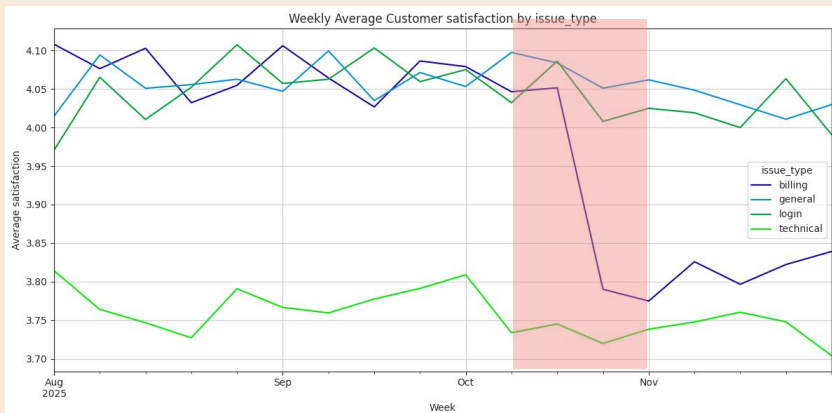
wait_time_missing	
satisfaction	
1	0.315789
2	0.087364
3	0.042002
4	0.039166
5	0.036459





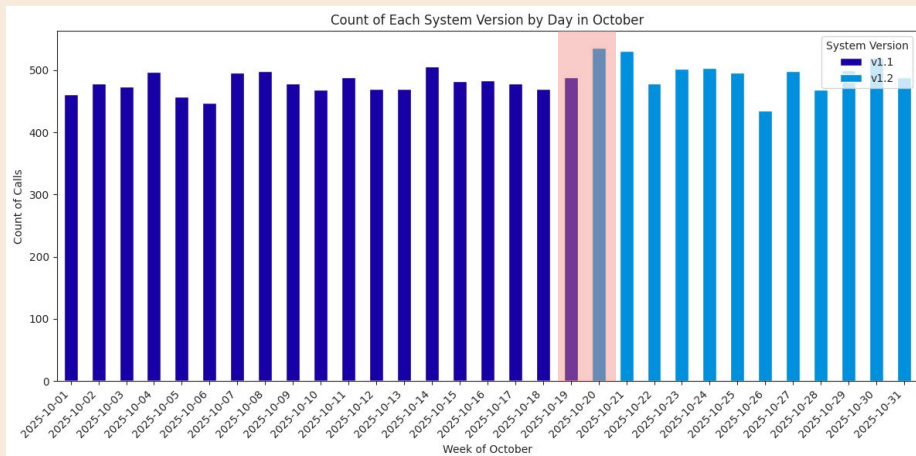
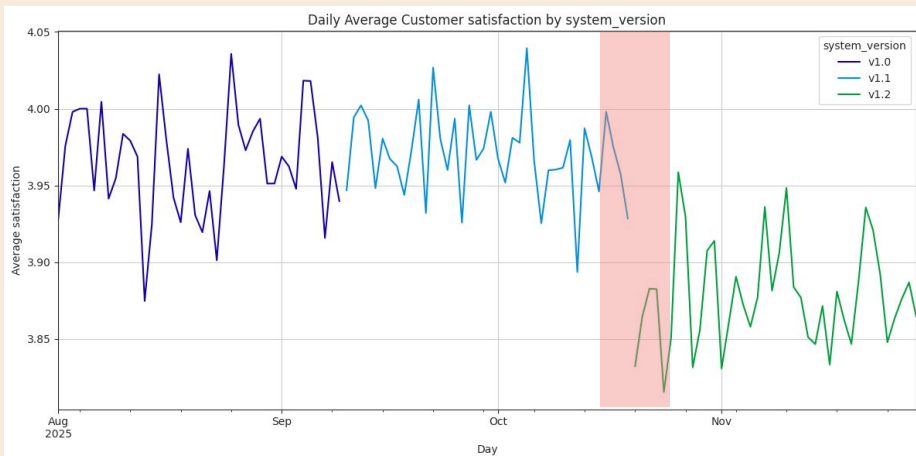
The Decline

- In mid-October, there's a significant dip in **satisfaction** for customers in **North** region, **Premium** customers and customers with **billing** issues
- Investigate!



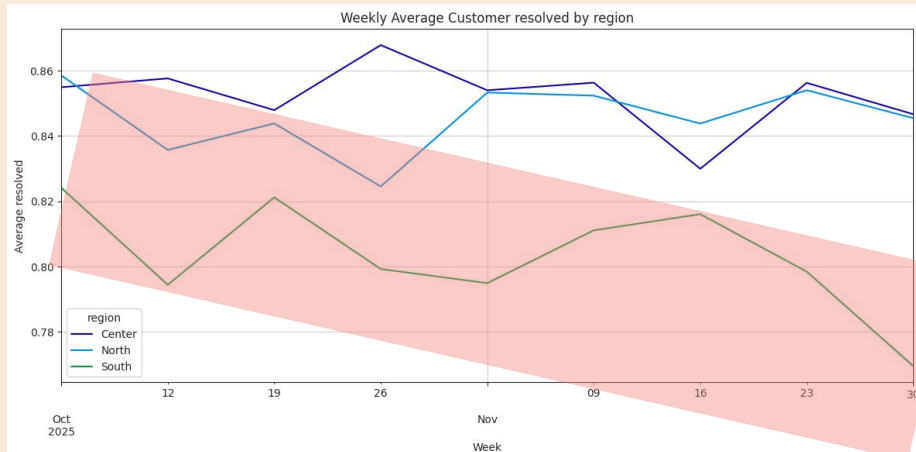
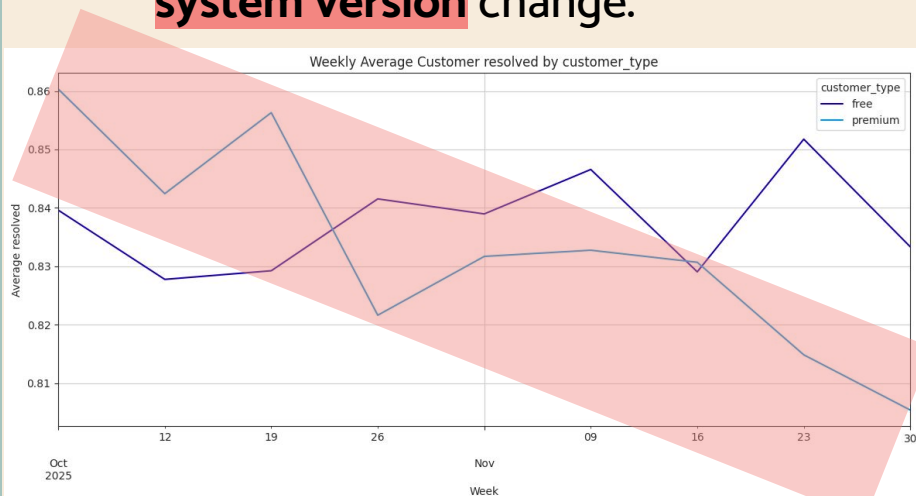
The Cause

- The **satisfaction** decline coincided with the introduction of a new **system version** (v1.2) on October 20th
- New version degraded the quality of service
- Quick fix: revert (rollback) system to previous version



↻ The Effect

- A clear positive correlation exists between **resolved** rates and **satisfaction** levels.
- The decrease in **resolved** rates was more pronounced among customers affected by the **system version** change.





Conclusion

- Revert to **system version** v1.1, review and improve deployment process to prevent future issues
- Improve the **resolved** rate and decrease customer **wait time** by implementing clearer Roles and Responsibilities (R&R) within the department, prioritizing **premium** customers. Approach with balance to avoid negatively impacting the conversion funnel.
- Verify these solutions by monitoring **satisfaction** and **resolved** rates of relevant customer segments.

